

# Passive Reconnaissance - OSINT

7CSEF006W

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## How to conduct OSINT

- ➡ This lab is not a complete OSINT guide but demonstrates basic activities using <https://cwscenario.uk/>.
- ➡ Your goal is to explore open-source intelligence methods and tools to gather information about the company.
- ➡ You are collecting information, not exploiting the website, to understand how the company may have been compromised.
- ➡ The gathered data will help you understand the company, its services, and potential weaknesses.
- ➡ This is an open-ended lab. You may use any OSINT tools, and a few examples are provided below:
  - ↳ OSINT requires time and persistence. The aim is to collect useful information for later analysis or testing.
  - ↳ A comprehensive list of resources: [Awesome OSINT Repository](#).
  - ↳ More tools: [OSINT Tools](#).
  - ↳ Tool collection: [OSINT Stuff](#).
  - ↳ An essential reference: [OSINT Framework](#).

## STOP and READ

- ➡ Before starting, ensure your lab environment is set up properly if using university devices:
  - ↳ Delete any existing VirtualBox VMs to start fresh.
  - ↳ Get the required VMs from **C:\VirtualMachines**. If outdated, download the latest version using the [lab environment guide](#).
  - ↳ Latest Kali version: **kali-linux-2024.1-UoW-virtualbox-amd64**.
  - ↳ The Kali VM is a .7z file. Extract it and add it manually instead of importing it as an OVA [See instructions](#).
  - ↳ Save the extracted VM files in **C:\VirtualMachines**.

### Requirements and Notes

- ➡ **NOTE** — Only your Kali Linux VM should have internet access.
- ➡ Required VM:
  - ↳ Kali Linux
- ➡ Ensure your Kali VM is connected to the internet using NAT.
- ⚠ This is **NOT** a hacking tutorial. You are fully responsible for your actions outside the lab.

## 1 Passive Reconnaissance: Open Source Intelligence (OSINT)



### Reconnaissance and Information Gathering phase

- ➡ Reconnaissance and information gathering are key stages in any penetration test or security audit.
- ➡ The more information you collect, the more options you have for finding and exploiting vulnerabilities.
- ➡ If internet access is required for installing tools or other tasks, it will be specified.
- ➡ OSINT involves using various techniques and tools to gather information online, often automating multiple activities.
- ➡ We will use some tools in this lab, but you are encouraged to explore others.
- ➡ For a detailed list of OSINT sources, see the [OSINT Framework](#).
- ➡ Use the domain <https://cwscenario.uk> for all OSINT investigations — it is safe and dedicated for testing.

## 1.1 TheHarvester

➡ Since [TheHarvester](#) searches for information already available online, your search will go unnoticed and will not raise any alarms.

➡ This is passive reconnaissance — you are not scanning or attacking the company's servers, only collecting public information.

➡ Check all available options:

```
➤ theHarvester -h
```

➡ We will focus on:

↳ -d → Specify the domain

↳ -b → Specify the data source (e.g., yahoo, linkedin, bing, twitter, etc.)

↳ -l → Set the number of results for sources that limit requests.

↳ Example search:

```
➤ theHarvester -d westminster.ac.uk -b yahoo -l 100
```

↳ To search all available sources:

```
➤ theHarvester -d https://cwscenario.uk/ -b all
```

↳ To save results in HTML format for later review:

```
➤ theHarvester -d https://cwscenario.uk/ -b all -f myresults.html
```

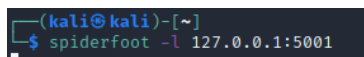
➡ For more details, visit the [TheHarvester GitHub repository](#).

## SpiderFoot

➡ SpiderFoot is another powerful OSINT tool, pre-installed on Kali Linux.

↳ Start SpiderFoot by running it on localhost:

```
➤ spiderfoot -l 127.0.0.1:5001
```



```
(kali㉿kali)-[~]  
$ spiderfoot -l 127.0.0.1:5001
```

**Figure 1.1:** Starting SpiderFoot on localhost port 5001

➡ While SpiderFoot is running, open a browser and go to 127.0.0.1:5001.

➡ SpiderFoot is module-based; some modules require API keys for better results. Without them, certain modules may not work.

- ➡ Click on “New Scan”, choose **Passive Scan**, give your scan a name, and specify either an IP address or a domain to investigate.
- ➡ Allow the scan to complete and explore the results.

The screenshot shows the SpiderFoot web interface. At the top, there's a navigation bar with links to Kali Furums, Kali NetHunter, Exploit-DB, Google Hacking DB, and OffSec. Below this is a 'New Scan' button. The main form has a 'Scan Name' field with 'firsttest' entered. The 'Scan Target' field has 'cwsenario.site' entered. To the right of the form, there's a list of supported target types: Domain Name, IPv4 Address, IPv6 Address, Hostname/Sub-domain, Subnet, Bitcoin Address, E-mail address, Phone Number, Human Name, Username, and Network ASN. Below the form, there are three radio buttons for scan type: 'All' (selected), 'Footprint', and 'Investigate'. The 'All' option is selected, and the 'Passive' option is also visible. At the bottom, there's a 'Run Scan Now' button.

**Figure 1.2:** Creating a new SpiderFoot scan on the testing domain

## SIGIT - Simple Information Gathering Toolkit

- ➡ SIGIT is another OSINT tool for gathering usernames, emails, and other information.

↳ Clone and install SIGIT:

```
➤ sudo git clone https://github.com/termuxhackers-id/SIGIT.git
```

```
➤ cd SIGIT
```

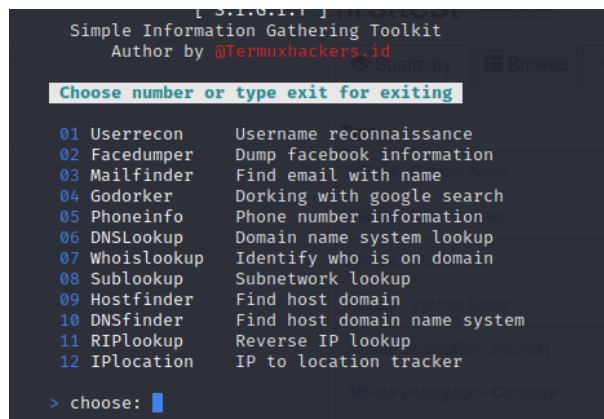
```
➤ chmod a+x install.sh
```

```
➤ sudo ./install.sh
```

↳ Run SIGIT:

```
➤ python3 sigit.py
```

- ➡ From the SIGIT interface, choose an option by entering its number (e.g., **01** for usernames).
- ➡ Try searching for a username, email, or domain to see what public data can be gathered.



**Figure 1.3:** SIGIT - Simple Information Gathering Toolkit

## Recon-ng

### Recon-ng Access Notice

- ➡ Recon-ng works only outside the university network unless connected via EduRoam.
- ➡ It is currently blocked on lab machines connected to the university network.

➡ Recon-ng is a full-featured framework for OSINT data gathering from various sources like Google, Twitter, and Shodan.

↳ Start Recon-ng:

```
>- recon-ng
```

↳ Install the Hackertarget module:

```
>- marketplace install hackertarget
```

↳ Search for the module:

```
>- modules search hackertarget
```

↳ Load the module:

```
>- modules load recon/domains-hosts/hackertarget
```

↳ Set the target source:

```
>- options set source https://cwscenario.uk/
```

↳ Run the module:

```
>- run
```

↳ To install all available marketplace modules:

```
➤- marketplace install all
```

↳ To list available modules:

```
➤- marketplace search
```

➡ Recon-ng can also be used alongside recon-web, which runs on port 5000 and provides a GUI for collected data.

➡ For tutorials, check:

↳ [Recon-ng Usage Example](#)

↳ [Recon-ng on SecurityTrails](#)

↳ [Hackersploit Video Playlist](#)

## Sock Puppet Accounts

➡ **Where to Collect Social Media Information:** To gather social media intelligence (SOCMINT), start by identifying relevant objectives and using multiple platforms.

➡ For a detailed guide, see [this article from Maltego](#).